

Assignment 5 – Abramov Aleksandr

Question 1. Explain in detail zero-shot, one-shot, and few-shot learning. Define each setup, give representative methods and benchmarks, analyze strengths and weaknesses, and state when each approach is preferable. (You may include brief literature).

1. Introduction

Training large neural networks generally requires large labelled datasets (a typical example would be ImageNet with millions of images and thousands of classes). However, labelled data is scarce in real-world problems, and collecting so many examples is either extremely expensive or impossible at all (rare diseases, facial features, etc.). The solution is to train a model using an existing large dataset (or, even better, take a known architecture with pretrained weights) and to devise a way to transfer its knowledge to new problems with minimal training using zero, one, or a few examples from the new task. Basically, one wants to create a neural network that can be easily adapted to new problems with a small set of examples.

A common generalization is the so-called N -way K -shot episodic learning. At test time, the model is given a set of N classes with K examples each (called a support set), and its objective is to classify a set of unseen examples (called a query set) from the same N classes. In these terms, zero-shot learning (ZSL) corresponds to $K = 0$, one-shot learning (OSL) to $K = 1$ and few-shot learning to $K > 1$ but generally small.

2. Zero-Shot Learning

2.1. Problem setup

In zero-shot scenarios, the model is required to recognize objects/classes for which no examples have been available during training. The applications include detecting new diseases, categorizing newly introduced products, and much more.

Formally, the label space is partitioned into $Y = Y_s \cup Y_u$: the seen classes Y_s (training labels available) and the unseen classes Y_u (new objects to be classified), and the task is to distinguish objects from Y_u while training only from Y_s . The general assumption is that $Y_s \cap Y_u = \emptyset$, although the condition may not always hold, since real-world problems generally do not limit inference to new classes but may also work with existing ones (for example, a model trained to classify clothing items is required to not only discover new classes but also to distinguish the known ones).

The general solution relies on the so-called semantic side information. Simply put, one requires two mapping functions $\theta(x): X \rightarrow \mathbb{R}^{d_2}$, $\varphi(y): Y \rightarrow \mathbb{R}^{d_1}$ and a compatibility rule $F(x, y): X \times Y \rightarrow \mathbb{R}$. The mapping functions are devoted to producing the embeddings corresponding to objects ($\theta(x)$) and class labels ($\varphi(y)$), which can be compared using $F(x, y)$. Those embeddings may be either hand-crafted descriptions (“can fly”, “has color pink”, etc.) or generated latent representations from large models, Word2Vec, knowledge graphs, etc. The classifier is then required to compute the embedding of the new object using θ and compare it to all registered class embeddings φ using F . Formally, the prediction $\hat{y}(x)$ for the new object is defined as follows:

$$\hat{y}(x) = \underset{y \in Y_c}{\operatorname{argmax}} F(\theta(x), \varphi(y))$$

where Y_c is the set of labels the model can output at inference

Generally, $Y_c = Y_u$ (only unseen classes can be produced as output), which is the conventional zero-shot learning scenario. Yet, $Y_c = Y$ (training classes can be produced as output as well) is a more realistic setting (generalized zero-shot learning), which requires careful handling of the model’s bias towards the seen classes. Nonetheless, the general idea of the approach persists.

So, the training procedure is devoted to tuning the parameters of θ , φ , and F (in fact, the mappings θ and φ may be fixed with only F left for training). Then, to add a new class into the setup for inference, one should just define its semantic embedding $\varphi(y)$, and the model should be able to work with it directly without additional training.

2.2. Representative methods

The representative methods of zero-shot learning include the following:

- DeViSE [1], which uses visual features from a pretrained CNN as θ , skip-gram word vectors as φ , and a hinge ranking function as F ;
- f-CLSWGAN [2], CADA-VAE [3], TF-VAEGAN [4], and others that use conditional generative adversarial networks or variational autoencoders to synthesise semantic embeddings for unseen classes, and train a traditional classifier on the unions of real and generated features;
- CLIP [5] and ALIGN [6] that rely on a shared image-text feature space, thus having both θ and φ in the same space, enabling their comparison using a simple cosine similarity.

2.3. Benchmark

A comprehensive benchmark of those and other methods has been presented in [7]. Since the task is basically a classification problem, all traditional ideas from that field are applicable here. Nonetheless, the paper introduces additional techniques to improve the robustness of the benchmarks.

First, a number of datasets have been developed specifically for the zero-shot scenario: aPY, AWA1, SUN, and CUB. The primary trait of those datasets is the semantic-oriented design: each dataset possesses several common shared attributes between classes that are perfect for zero-shot classification. Moreover, the classes are specifically designed to mix different classes, especially between the ones that were used for training and the ones that were not.

Second, traditional accuracy has limited applicability in zero-shot learning scenarios due to a high number of classes with a significant imbalance, which is especially crucial in the generalized ZSL. In the problem at hand, one is more interested in having high performance on sparsely populated classes rather than a general dataset-wide accuracy. Thus, a common metric in zero-shot learning is the so-called average per-class accuracy: the correct predictions are averaged independently within each class, and the cumulative sum is then divided by the number of classes. For the generalized ZSL, they additionally compute the harmonic mean between the training and test accuracies to account for the model’s inherent bias towards the classes seen during training, aiming for high accuracy on both seen and unseen classes.

For the specific accuracy values, please refer to the original study.

2.4. Strengths and Weaknesses

The primary advantage of the approach is the simplicity of adding new classes. There is virtually no cost to adding as many classes to the annotations as desired. Additionally, zero-shot learning naturally combines different types of data (i.e., image and text) and only requires an embedding mapper for each of them to operate, which is useful for models like VLM, which are surprisingly strong out of the box. Finally, some zero-shot techniques are extremely interpretable if the embeddings are semantically valid and reflect real attributes of relevant objects.

The main issue in zero-shot learning is the dependence on mapper quality: if the semantic embeddings are noisy, no good classification is to be expected for new classes. This leads to zero-shot techniques usually being very sensitive and prone to identifying false correlations: models are highly biased towards representations of seen classes, which become “hubs” for many objects, and unseen classes drift away in the feature space. Although the issue can be solved using advanced loss functions and calibration, this remains a challenge in the approach.

3. One-Shot Learning

3.1. Problem setup

In one-shot scenarios, the model is required to recognize novel classes from exactly one labelled example per class ($K = 1$). Typical applications include signature or fingerprint verification, face recognition, and more.

Formally, the objective of the model at inference is to classify a query set $Q = \{x_j\}$ into N classes given a support set $S = \{(x_i, y_i)\}_{i=1}^N$ with one example for each of those classes. Notably, this is not the traditional classification task but rather a meta-learning problem: the model is required to not just assign the correct labels, but also to account for the additional context (the support set) specified with each request.

The general solution is similar to the zero-shot scenario, with the distinction that one does not create a class embedding $\varphi(y)$ but uses the corresponding example from the support set as the representation for that class. One can view this as a special case of the zero-shot setup with $\varphi(y) = \theta(x_i)$: the class embedding being the embedding of its single support example. Thus, the prediction $\hat{y}(x)$ for the new object is defined as follows:

$$\hat{y}(x) = \underset{i \in [1; N]}{\operatorname{argmax}} F(\theta(x), \theta(x_i))$$

where θ is the trained mapping producing embeddings for objects at hand, and F is the trained similarity function that enables comparison between objects in the representation space.

Training typically follows an episodic protocol: each iteration is given an N -way 1-shot episode (a query set and a support set for N classes) sampled from a training dataset with classes C_{train} , which is used to make a step of gradient descent and update the model's weights. For evaluation, the objects are sampled from another, disjoint class set C_{test} , which does not intersect with the training classes ($C_{train} \cap C_{test} = \emptyset$). In this setup, the goal of the model becomes not to recognize the objects but to learn how to compare them with class examples: the trained embedding space θ should generalize well from the training classes to new ones.

3.2. Representative methods

The simplest implementation of one-shot learning may be simply reusing the embedding computation techniques from zero-shot, since those are readily applicable to the task as well. Moreover, those usually serve as a good baseline paired with a nearest-neighbour search in representation space.

Nonetheless, several classical implementations specialized for one-shot learning have been presented in the literature:

- Siamese networks [8], which employ two identical branches with tied weights that map pairs of objects into embeddings and employ a learned head to predict whether two inputs belong to the same class. In inference, the support example with the highest predicted similarity is considered the output of the model;
- Matching networks [9] that introduce an attention-weighted nearest-neighbour classifier to generate class prototypes in the representation space and classify new examples according to the distance to those prototypes;
- Memory-augmented neural networks [10] that include an external memory module inspired by Neural Turing Machines, allowing the model to store a fresh example and use it for prediction.

3.3. Benchmark

The canonical one-shot benchmark is Omniglot [11] – a dataset of 1623 handwritten characters from 50 alphabets with 20 examples per class. It was deliberately designed to have many classes with few examples for each, which makes it a natural fit for one-shot evaluation. Beyond that, miniImageNet and tieredImageNet, subsets of the original ImageNet, are also standard datasets for one-shot benchmarks that force large semantic gaps between seen and novel classes.

A comprehensive progress report on the Omniglot dataset, surveying both algorithmic and human performance, has been presented in [12]. It revisits the original challenge of one-shot learning, compares various deep learning techniques with prior probabilistic methods and human baselines, and reports the accuracy in different setups. Similar to zero-shot, the metric of interest is the average per-class accuracy that efficiently handles class imbalance and natural bias, which are also inherent to one-shot techniques.

3.4. Strengths and Weaknesses

The main advantage of one-shot learning is the minimal labelling cost for new classes: a single example per class is enough, which is invaluable in domains where collecting more references is impossible or prohibitively expensive. Inference is also cheap — most metric-based methods require only a single forward pass through the embedding network, with no gradient updates at test time. Moreover, a single embedding network (compared to two in zero-shot) is advantageous in terms of robustness and stability of the approach: making one good embedding network is definitely easier than making two.

The primary weakness of the methodology is high variance: a single support example may be atypical for the class, pulling the prototype away from the natural placement of the class in the representation space. As a result, one-shot accuracy may fluctuate heavily across episodes and across choices of the support

example. Similar to zero-shot, the approach also makes a strong assumption that the seen and new classes share a useful embedding geometry, which might not be the case in reality. The choice of the similarity function and the embedding backbone is crucial: a poor choice can drastically degrade the performance.

4. Few-Shot Learning

4.1. Problem setup

Few-shot learning generalizes the one-shot approach to a usually small number $K > 1$ of labelled examples per novel class, which are generally available for various medical imaging tasks, defect detection, recommender systems, and more. In each episode, the model is given a support set of N classes with K examples for each, and a query set of unlabelled examples to classify. The training and evaluation procedures are otherwise identical to the one-shot case: episodic sampling from a training set for tuning the parameters, and evaluation using a disjoint novel class set.

The presence of multiple support examples per class allows averaging out the noise of any individual one to create a more robust embedding representation, which attempts to solve the weaknesses of the one-shot approach. A typical implementation averages the support examples into a class prototype and performs a nearest-neighbor classification using the obtained embeddings. In fact, this is again a special case of the zero-shot framework with $\varphi(y)$ computed from the support examples of that class.

$$\hat{y}(x) = \underset{i \in [1; N]}{\operatorname{argmax}} F\left(\theta(x), \frac{1}{K} \sum_{x', y' \in S_k} \theta(x')\right)$$

4.2. Representative methods

With few-shot learning being arguably the most common approach of the three, there are a number of its implementations from varying perspectives, which are usually separated into several families.

- Metric-based techniques that learn an embedding space where simple distance metrics would suffice for the nearest-neighbor classification. These include prototypical networks [13], which compute the class prototype as the mean of its supporting examples, and relation networks [14] that replace the fixed Euclidean metric with a learned comparator on the resulting embeddings.
- Optimization-based methods aiming to learn an initialization that adapts fast to the support set. The primary example here is MAML [15] – a model-agnostic meta-learning framework that finds an initialization such that a small number of gradient descent steps on the support set produce a strong classifier on the query set. Variants include Reptile [16], which uses a first-order approximation to avoid

differentiation through the inner loop, and ANIL [17], which freezes the embedding backbone and adapts only the classifier head.

- Model-based approaches that look for architectures with internal adaptation mechanisms. The primary examples are the memory-augmented neural networks mentioned in the one-shot section, which are also applicable in the few-shot scenario.
- Transfer baselines, which optimize for a strong pretrained backbone with a small head that can be easily finetuned to the new class. Numerous studies show that pretraining a backbone with a plain supervised loss on a huge dataset reveals interesting patterns in the data that can be efficiently transferred to new tasks by finetuning a linear classifier head on the support set.
- In-context learning attempts to pack the support set fully into the prompt of a large pretrained model to readily get the result. This approach is especially popular in NLP, where modern large language models show great results in solving new tasks by reading a handful of demonstrations in their prompts [18].

4.3. Benchmark

A comprehensive benchmark and re-evaluation of few-shot methods has been presented in [19]. The study fixes a backbone, an episodic protocol, and a set of standard datasets, and runs major techniques for comparison. Its main finding is that simple pre-training baselines often match or exceed specialized meta-learning models, which is a key result that has reshaped the family of few-shot techniques.

The classic benchmarks are again the miniImageNet and tieredImageNet datasets, while Meta-Dataset [20] is another interesting choice that aggregates ten heterogeneous datasets, deliberately stressing cross-domain knowledge transfer. The evaluations also typically report the average per-class accuracy as the most robust metric for the task.

4.4. Strengths and Weaknesses

Few-shot learning is arguably the most practical of the three regimes: a dozen samples per class is usually achievable in most domains, and the variance of few-shot methods is generally much lower compared to one-shot and zero-shot techniques. Moreover, modern pre-trained backbones combine well with a tiny linear head on the support set and often demonstrate prevalent results.

The main weakness remains the distribution shift between training and testing classes, especially across different fields in cross-domain problems. Additionally, few-shot techniques usually have higher computational costs, and plausible optimizations are sometimes unstable and do not give the expected performance improvement. Furthermore, the relative abundance of data may lead to underutilization, making traditional supervised training a better alternative despite the costs.

5. Discussion

The three setups sound very similar in their original meta-learning formulations, yet lead to completely different solutions with interesting side effects of each technique. Zero-shot learning relies on the semantic meanings of classes and objects and is preferable when no labels at all can be obtained for the novel classes, but reliable attribute descriptions can be devised. It is also the natural choice when the class set changes constantly, as in search or recommendation. One-shot learning leans on the embedding geometry: a single canonical reference must already lie close to the true class manifold. This approach fits problems like signature verification and face recognition, where a neural network must distinguish pairs of objects (i.e., people) from just one sample. Finally, few-shot learning is the most general and practical regime, well-suited for tasks with a handful of examples per class: medical imaging with a few confirmed cases per condition, industrial defect detection, and personalisation tasks.

The main disadvantage of all three techniques is that representation quality is the bottleneck: a frozen pretrained backbone producing good features with a tiny adaptation head usually beats even the best meta-learners trained from scratch. Additionally, all methods are prone to the seen-class bias, which makes the evaluation fragile and often misleading. Future progress is therefore likely to come less from new architectures and more from better pretraining, calibration, and honest cross-domain evaluation.

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